ViAmpleHate: A Proposed AmpleHate- and PhoBERT-based Approach for Vietnamese Hate Speech Detection

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Abstract

Detecting hate speech on Vietnamese social media is difficult because hateful intent is rarely carried by explicit named entities; instead it hides in informal group references, slang, sarcasm, and indirect insinuation. Target-aware models such as AmpleHate have shown that attending to the relationship between an utterance and the target it addresses helps detect even implicit hate, yet AmpleHate's original pipeline relies on English named-entity recognition (NER) and English-oriented target categories, so it transfers poorly to Vietnamese: on our training data English NER finds a usable target in only about 0.09% of comments, collapsing the model into a plain sentence classifier. We propose **ViAmpleHate**, a Vietnamese adaptation of AmpleHate built on PhoBERT. ViAmpleHate (i) replaces English NER with Vietnamese NER and a curated *target-cue* bank, raising target coverage from ≈0.09% to ≈19–45% depending on the dataset; (ii) adds a separate *attack-cue* bank and models three relation channels—explicit target, implicit context, and attack—through a *relation-bank attention*; (iii) replaces fixed-scalar injection with an *instance-adaptive gate*; and (iv) trains with weighted cross-entropy combined with a contrastive objective. On ViHSD and VOZ-HSD in a binary NON-HATE/HATE setting, ViAmpleHate improves macro-F1 and the minority HATE-class F1 over an AmpleHate baseline and over TF-IDF, BiLSTM, and PhoBERT-CNN baselines, with the largest and clearest gains on the minority class (HATE-F1 $+0.0508$ on VOZ-HSD); improvements on ViHSD are small and we report them as preliminary, pending a controlled, multi-seed evaluation. We further analyse how target-cue coverage relates to these gains, supporting—rather than proving—the view that target-aware modeling should be adapted to Vietnamese rather than transferred verbatim from English.

1 Introduction

The rapid growth of Vietnamese-language social media has been accompanied by a corresponding spread of hateful and abusive content. Such content harms targeted individuals and communities, degrades online discourse, and creates legal and reputational risk for platforms. Because the volume of user-generated text vastly exceeds what human moderators can review, automatic hate speech detection has become a practical necessity. Yet most progress has centred on English and other high-resource languages, and methods that work well there often degrade when applied to Vietnamese.

Vietnamese hate speech detection is hard for three intertwined reasons. First, social-media language is extremely noisy: users abbreviate, write in *teencode*, elongate characters for emphasis, embed emojis with pragmatic meaning, and—deliberately or not—misspell words, including obfuscating profanity to evade filters. Second, and most importantly for this work, the *target* of an attack is rarely a named entity. Whereas English hate speech frequently names a person, organization, or nationality, Vietnamese hostility is typically aimed at a group through pronouns, kinship or group nouns, regional labels, or informal forms of address. Third, the data is severely imbalanced: in both datasets we study, the HATE class accounts for only about one tenth of all comments, so a model can reach high accuracy while detecting almost no hate.

A promising line of work models hate speech in a *target-aware* manner. AmpleHate (Lee et al., 2025), in particular, shows that explicitly attending to the relationship between a sentence and its potential target improves detection of *implicit* hate, where hostility is conveyed without overt slurs. AmpleHate extracts candidate targets with NER, computes a target-aware attention vector, and injects it into the sentence representation before classification. This is attractive for Vietnamese, where the target is so central—but the original pipeline is built around English NER and English target categories. When we run it on Vietnamese with an English NER component, it finds a usable target in only 21 of 24,048 training comments (about 0.09%). For the overwhelming majority of inputs the model falls back to the global `[CLS]` representation and behaves like an ordinary PhoBERT classifier, discarding precisely the target-aware signal that motivated it.

We propose **ViAmpleHate**, a Vietnamese adaptation of AmpleHate that re-engineers each stage of the pipeline around how Vietnamese speakers actually express hate. Our contributions are:

1. **Vietnamese target extraction.** We replace English NER with a Vietnamese NER model and a curated *target-cue* bank of pronouns, group nouns, and informal address forms, raising the fraction of comments with a detected target from ≈0.09% to ≈18.8–20.0%.

2. **Separate target and attack channels.** We add an *attack-cue* bank for hostile predicates and model target and attack cues as distinct signals through a three-channel **relation-bank attention**.

3. **Adaptive fusion (plus an implementation note).** We replace AmpleHate's fixed-scalar injection with an **instance-adaptive gate**. We additionally describe an implementation-level correction to a batch-level attention computation in our port, which we report as an implementation note rather than a primary contribution.

4. **Training objective.** We combine weighted cross-entropy with a contrastive loss that sharpens HATE/NON-HATE separation under imbalance.

5. **Empirical study.** We evaluate on ViHSD and VOZ-HSD against five baselines and analyse where and why the gains arise, including a target-coverage analysis and a qualitative study of remaining errors.

The broader message is that target-awareness is a genuinely useful inductive bias for hate speech detection, but only when adapted to the linguistic surface of the target language instead of being transferred verbatim from English.

2 Related Work

2.1 Hate speech detection and implicit hate

Automatic hate speech detection has evolved from lexicon- and feature-based classifiers (Davidson et al., 2017; Schmidt and Wiegand, 2017) to deep sequence models and, more recently, pretrained transformers. A persistent difficulty is *implicit* hate: messages that convey hostility without explicit slurs, relying on sarcasm, stereotypes, coded references, or in-group knowledge (ElSherief et al., 2021). Implicit hate is often more common than overt slurs and substantially harder for keyword-driven systems, motivating approaches that reason about *who* is targeted and *how*.

2.2 AmpleHate

AmpleHate (Lee et al., 2025) addresses implicit hate by amplifying the attention between a sentence's global representation and its candidate targets. It extracts target tokens with NER, computes a HeadAttention vector whose query derives from `[CLS]` and whose keys/values derive from target tokens, and injects the result with a fixed coefficient before classification. Its central insight—that the target is a load-bearing signal—directly motivates our work. Its limitations for Vietnamese are equally direct: it presumes English-style NER and target categories, injects relation information at a fixed strength for every instance, and models a single relation (target) while ignoring the hostile predicate, or *attack*, that turns a mention into hate.

2.3 Vietnamese hate speech detection

Vietnamese has seen growing interest in hate speech resources and models. ViHSD (Luu et al., 2021) provides a large-scale three-way (CLEAN/OFFENSIVE/HATE) dataset and is a standard benchmark. ViTHSD (Vo et al., 2025) reframes the task around *targets*, annotating the degree of hatred directed at each target; this reinforces the centrality of the target in Vietnamese hate speech and supports our target-aware design. Unlike ViTHSD, ViAmpleHate requires no span-level target annotation; it locates target and attack positions with NER plus cue banks, which keeps it applicable to datasets carrying only sentence-level labels.

2.4 PhoBERT and Vietnamese pretrained models

PhoBERT (Nguyen and Nguyen, 2020) is a RoBERTa-style (Liu et al., 2019) model—itself a robustly optimized variant of BERT (Devlin et al., 2019), which builds on the Transformer architecture (Vaswani et al., 2017)—pretrained on large Vietnamese corpora and is the de facto encoder for Vietnamese text classification. Because PhoBERT is trained on *word-segmented* text, word segmentation (e.g., with VnCoreNLP (Vu et al., 2018)) precedes tokenization. Hybrid architectures such as PhoBERT-CNN (Tran et al., 2023) add local feature extractors on top of contextual embeddings. We use PhoBERT-base as the shared encoder for both baseline and proposed models so that differences are attributable to target-aware modeling rather than the backbone.

2.5 Contrastive learning for text classification

Contrastive objectives encourage same-class representations to be close and different-class representations to be far apart, improving robustness and class separation, particularly under imbalance (Khosla et al., 2020). We adopt a contrastive auxiliary objective on the post-gate representation alongside weighted cross-entropy—a pairwise cosine-margin term inspired by supervised contrastive learning, not the exact SupCon loss—to tighten the boundary between the frequent NON-HATE class and the rare HATE class.

3 Datasets

3.1 ViHSD

ViHSD (Luu et al., 2021) is a Vietnamese hate speech dataset of social-media comments annotated with three original labels (CLEAN, OFFENSIVE, HATE) and pre-split into train/validation/test. After the binary relabeling of Section 3.3, the per-split distribution is in Table 1.

3.2 VOZ-HSD

VOZ-HSD (Thanh Nguyen, 2024b), released with ViHateT5 (Thanh Nguyen, 2024a), consists of comments from the VOZ forum (sourced from the public `tarudesu/VOZ-HSD` release). As with ViHSD, the data is split into train/validation/test, and its post-relabeling distribution is in Table 1. As a third-party resource, its detailed annotation procedure and licensing are documented by the dataset authors.

3.3 Binary relabeling and preprocessing

The only modification we make is *relabeling*; we do not alter or subsample the number of examples. We cast the task as binary classification by merging CLEAN and OFFENSIVE into NON-HATE and keeping HATE as the positive class. This focuses the problem on group-directed hate, as opposed to profanity or aggression in general, and yields a consistent label space across datasets. Each comment is then normalized (lowercasing; URL/symbol removal; collapsing elongated runs; teencode normalization; mapping frequent emojis to coarse pragmatic tags such as mockery, anger, disgust, or laughter), word-segmented—mandatory because PhoBERT is pretrained on word-segmented Vietnamese—and tokenized with the PhoBERT tokenizer. The resulting class distribution is summarized in Table 1 and illustrated in Figure 1.

3.4 Cue bank construction

The target-cue and attack-cue banks are central to ViAmpleHate. Both are seeded manually from inspection of the training split and from common Vietnamese referential and offensive expressions, then normalized to match the preprocessing pipeline. The **target-cue bank** contains referential expressions that often introduce a person or group—pronouns, kinship and group nouns, regional and demographic labels, and informal address forms—but that are *not* by themselves indicators of hate. The **attack-cue bank** contains hostile predicates: insults, dehumanizing terms, threats, and strongly negative evaluations. Keeping the banks separate is deliberate: a target mention without a hostile predicate is usually not hate, and a hostile predicate without a clear group target is often mere offensiveness. Each cue is normalized, word-segmented, and tokenized, then matched against the token stream by full token-sequence matching so a multi-token Vietnamese phrase is matched as a unit rather than via a stray subtoken. The banks are deliberately small, hand-curated lexicons—16 target cues and 24 attack cues (Appendix B)—seeded from inspection of the training split and common Vietnamese expressions. Because they are seeded from training data they risk encoding train-specific patterns, and their size bounds recall on novel slang; we report the resulting coverage in Section 6.3.

Dataset	Split	NON-HATE	HATE	Total
ViHSD	Train	21,492	2,556	24,048
ViHSD	Dev	2,402	270	2,672
ViHSD	Test	5,992	688	6,680
VOZ	Train	26,993	3,007	30,000
VOZ	Dev	4,520	480	5,000
VOZ	Test	4,487	513	5,000

Table 1: Dataset statistics after binary relabeling. HATE is ≈10% throughout, so macro-F1/HATE-F1 matter more than accuracy.

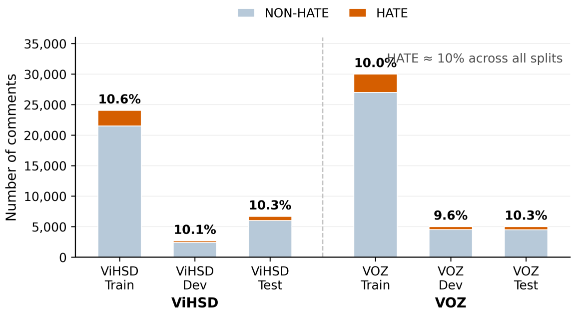

Figure 1: Class distribution of ViHSD and VOZ-HSD after binary relabeling.

4 Methodology

4.1 Problem formulation

Given a Vietnamese comment x, the model predicts $y \in \{\text{NON-HATE}, \text{HATE}\}$. ViAmpleHate inherits AmpleHate's target-aware idea and adapts it through five changes: Vietnamese target extraction, separate target/attack cue channels, relation-bank attention, a corrected batched attention computation, and adaptive-gate injection (Figure 2).

4.2 Text preprocessing

As above, comments are normalized, word-segmented, tokenized with the PhoBERT tokenizer, and truncated/padded to $\text{max\_len} = 256$.

4.3 Multi-signal target and attack extraction

The target index set combines Vietnamese NER and target-cue matching; the attack index set comes from attack-cue matching. If either is empty, the model falls back to `[CLS]` so an implicit, sentence-level representation is always available:

$$\begin{aligned}
T_x &= M_{\text{NER}}(x) \cup M_{\text{target}}(x),\\
A_x &= M_{\text{attack}}(x),\\
T_x &\leftarrow \{0\} \text{ if } T_x = \varnothing,\\
A_x &\leftarrow \{0\} \text{ if } A_x = \varnothing.
\end{aligned} \tag{1}$$

4.4 PhoBERT encoder

$$\begin{aligned}
H &= \text{PhoBERT}(x)\\
&= [\,h_0, h_1, \ldots, h_n\,], \quad h_i \in \mathbb{R}^d,
\end{aligned} \tag{2}$$

with $d = 768$; h_0 is the `[CLS]` representation (global context), and the target/attack positions provide localized evidence.

4.5 Relation-bank attention

We build three relation views—explicit target, implicit context, and attack:

$$\begin{aligned}
r_{\text{exp}} &= \text{HeadAttn}(h_0, H[T_x]),\\
r_{\text{imp}} &= \text{HeadAttn}(h_0, h_0),\\
r_{\text{atk}} &= \text{HeadAttn}(h_0, H[A_x]),
\end{aligned} \tag{3}$$

where each module over $E \in \mathbb{R}^{m \times d}$ computes

$$\alpha = \text{softmax}\!\left(\frac{(W_q h_0)(W_k E)^\top}{\sqrt{d}}\right), \quad r = \alpha\,(W_v E). \tag{4}$$

The three vectors are fused as $r = W_r[\,r_{\text{exp}}; r_{\text{imp}}; r_{\text{atk}}\,] + b_r$.

4.6 Batched attention

In our re-implementation, an AmpleHate-style attention computed as $QK^\top$ over the whole batch ($Q, K \in \mathbb{R}^{B \times d} \Rightarrow QK^\top \in \mathbb{R}^{B \times B}$) can mix information across samples. We therefore use batched attention so each sample attends only to its own tokens, with $Q \in \mathbb{R}^{B \times 1 \times d}$, $K \in \mathbb{R}^{B \times m \times d}$:

$$\text{scores} = \text{bmm}(Q, K^\top)/\sqrt{d} \in \mathbb{R}^{B \times 1 \times m}, \tag{5}$$

with padding masked before softmax ($\text{scores}_j = -\infty$ if $\text{mask}_j = 0$). We treat this as a correction to our own port rather than a claim about the original AmpleHate, and do not evaluate it in isolation.

4.7 Instance-adaptive relation gate

We replace the fixed scalar with a per-sample gate:

$$g = \sigma\big(W_g[\,h_0; r\,] + b_g\big), \qquad z = h_0 + g \cdot r. \tag{6}$$

With clear evidence, the model raises g and relies more on the relation vector; with weak or absent cues, it lowers g and leans on the global representation. The fused z passes through dropout and a linear layer to produce logits.

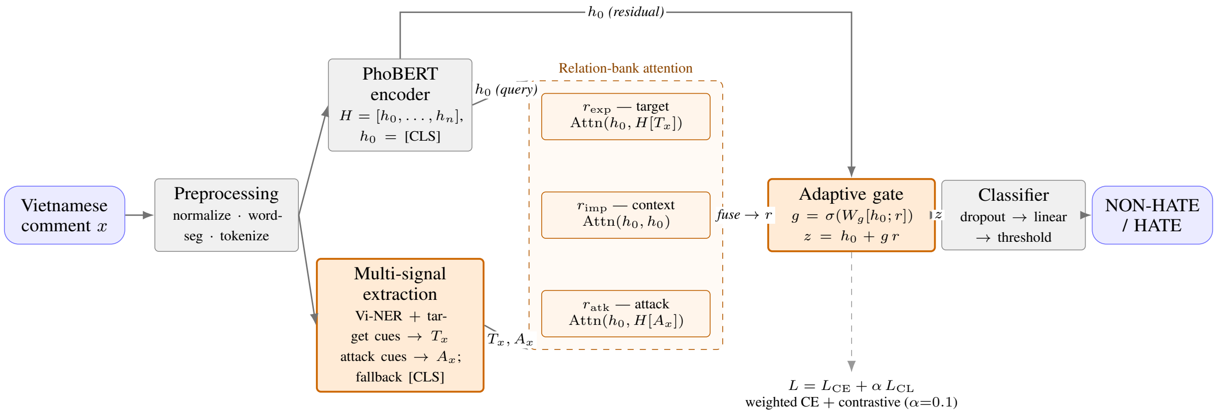

Figure 2: Overall architecture of ViAmpleHate. The comment is normalized, word-segmented, and encoded by PhoBERT into hidden states $H = [h_0, \dots, h_n]$, with h_0 the [CLS] (global) representation. Vietnamese NER and a target-cue bank produce the target index set T_x, while an attack-cue bank produces A_x (falling back to [CLS] when a set is empty). A relation-bank attention with three per-sample channels—explicit target (r_{exp}), implicit context (r_{imp}), and attack (r_{atk})—is fused into a single relation vector r and injected into h_0 through an instance-adaptive gate ($g = \sigma(W_g[h_0; r])$, $z = h_0 + g\,r$). A linear classifier with a validation-tuned threshold yields the NON-HATE/HATE decision; training minimizes $L = L_{\text{CE}} + \alpha L_{\text{CL}}$.

4.8 Training objective

We train with $L = L_{\text{CE}} + \alpha\, L_{\text{CL}}$, $\alpha = 0.1$. Weighted cross-entropy $L_{\text{CE}} = -\sum_c w_c y_c \log \hat{y}_c$ handles imbalance by up-weighting HATE, with label smoothing (Müller et al., 2019) (the class weights w_c, the label-smoothing value, and the margin m below are reported in Appendix A). The contrastive term on z (with $s_{ij} = \cos(z_i, z_j)$) pulls same-class pairs together and pushes different-class pairs apart:

$$L_{\text{CL}} = \frac{1}{N} \sum_{i \neq j} \big[\mathbb{1}[y_i{=}y_j](1 - s_{ij}) + \mathbb{1}[y_i{\neq}y_j] \max(0, s_{ij} - m) \big]. \tag{7}$$

4.9 Inference and threshold selection

At inference we apply a decision threshold chosen on validation by maximizing macro-F1 and then fixed for test:

$$t^* = \arg\max_t \ \text{MacroF1}\big(y,\ \mathbb{1}[p_{\text{HATE}} \geq t]\big). \tag{8}$$

4.10 Computational considerations

The added components are lightweight relative to PhoBERT. The three HeadAttention modules and the gate operate on the pooled `[CLS]` query and a small number of cue tokens per sample, so the extra cost is a few linear projections; the corrected batched attention removes the spurious $B \times B$ interaction and is cheaper at training time. Cue matching is performed once during preprocessing.

5 Experiments

5.1 Baselines

We compare against five baselines, all binary. **TF-IDF + LR** and **TF-IDF + SVM** use sparse lexical features. **BiLSTM + fastText** (Bojanowski et al., 2017) combines static embeddings with a recurrent (LSTM) model (Hochreiter and Schmidhuber, 1997). **PhoBERT-CNN** (Tran et al., 2023) stacks CNN feature extractors on PhoBERT. **AmpleHate-PhoBERT** is a port of the original with English NER, a single HeadAttention, and fixed injection $z = h_0 + e\, r_{\text{base}}$ ($e = 1.0$). It shares the PhoBERT-base encoder. As Table 2 makes explicit, each model is run at its own intended configuration—ViAmpleHate deliberately uses a longer context window (256) and a larger effective batch, which are part of the proposed system—so the comparison is between the full proposed system and the baseline as configured, rather than an isolation of individual components.

5.2 Implementation details

The encoder is `vinai/phobert-base` (768-d); Vietnamese NER is `NlpHUST/ner-vietnamese-electra-base`. Max length is 256. Learning rates are 2e−5 (encoder) and 5e−5 (head); dropout 0.1. Effective batch is 32 (16 × grad-accum 2). We train up to eight epochs and select the best checkpoint by validation macro-F1. $\alpha = 0.1$, and the threshold is selected on validation by macro-F1. Full

Component	Baseline AmpleHate-PhoBERT	ViAmpleHate-PhoBERT (proposed)
Encoder	PhoBERT-base	PhoBERT-base
Target extraction	English NER	Vietnamese NER + target-cue bank
Target coverage	≈0.09% of train (21/24,048)	≈18.8–20.0% via Vietnamese cues
Attack signal	Not modeled	Separate attack-cue bank
Attention	One HeadAttention	Three channels: target / implicit / attack
Attention computation	Batch-level `matmul` (mixes samples)	Per-sample `bmm` + masking
Fusion	Fixed injection $h_0 + e\,r$ (e=1.0)	Relation bank + adaptive gate $h_0 + g\,r$
Loss	Weighted CE	Weighted CE + contrastive (α=0.1)
Max length	128	256
Batch	16	16 $\times$ grad-accum 2 (eff. 32)
NER at evaluation	Off (⇒ `[CLS]` fallback)	On, consistent across splits

Table 2: Architecture and configuration comparison between the baseline and the proposed model.

hyperparameters are in Appendix A.

In terms of training dynamics, Figure 3 shows that ViAmpleHate converges stably on ViHSD, reaching its best validation macro-F1 at epoch 4.

5.3 Evaluation metrics

Our primary metrics are macro-F1 and HATE-class F1, which reflect minority-class performance far better than accuracy: under ≈10% positives, a trivial majority predictor already exceeds 0.89 accuracy while detecting no hate. Macro-F1 averages per-class F1; HATE-F1 isolates the class of interest.

5.4 Experimental design

Beyond final scores, our experiments test whether each modification addresses a concrete baseline limitation: Vietnamese NER and target cues target low coverage; the attack-cue bank addresses missing hostility modeling; relation-bank attention separates target, context, and attack; corrected batched attention removes cross-sample leakage; the adaptive gate replaces fixed injection; and the contrastive loss improves class separation. Section 6.3 examines target coverage directly.

6 Results and Error Analysis

6.1 Main results

Each model is reported at its own intended configuration (Section 3.3, Table 2); the proposed-vs-baseline rows therefore compare the full proposed system against the baseline as configured.

Transformer-based models clearly outperform the lexical and static-embedding baselines on the metrics that matter (Tables 3, 4), confirming that contextual representations are important when hate depends on context, informal phrasing, and the interaction between a mention and a hostile predicate. Within the transformer family, the AmpleHate

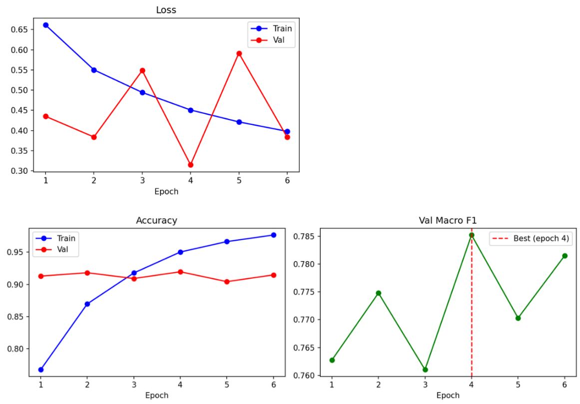

Figure 3: Training/validation curves of ViAmpleHate on ViHSD (best epoch 4, val macro-F1 = 0.7852). `training_curves_viamplehate.png`.

Model	Acc.	Mac-F1	HATE-F1
TF-IDF + LR	0.8910	0.7393	0.5404
TF-IDF + SVM	0.9126	0.7131	0.4739
BiLSTM + fastText	0.8454	0.7072	0.5060
PhoBERT-CNN	0.8945	0.7571	0.5745
AmpleHate (baseline)	0.9175	0.7792	0.6045
ViAmpleHate	**0.9205**	**0.7819**	**0.6081**
Δ *base*	*+.0030*	*+.0027*	*+.0036*

Table 3: Results on ViHSD (test). Best per column in bold (accuracy shown for reference).

Model	Acc.	Mac-F1	HATE-F1
TF-IDF + LR	0.9453	0.7745	0.5783
TF-IDF + SVM	0.9641	0.7831	0.5850
BiLSTM + fastText	0.8650	0.6712	0.4187
PhoBERT-CNN	0.9623	0.8150	0.6500
AmpleHate (baseline)	0.9643	0.8185	0.6557
ViAmpleHate	0.9420	**0.8371**	**0.7065**
Δ *base*	*–.0223*	*+.0186*	*+.0508*

Table 4: Results on VOZ-HSD (test). Best per column in bold among models (accuracy shown for reference; the baseline has the highest accuracy).

baseline improves over a plain PhoBERT classifier through target-aware attention, but its benefit is capped because English NER rarely detects valid Vietnamese targets, so most inputs revert to `[CLS]`. We compare only against models we trained under a common protocol; a comparison to published ViHSD results is left to future work.

ViAmpleHate improves macro-F1 and HATE-F1 on VOZ-HSD by a clear margin (HATE-F1 $+0.0508$). On ViHSD the differences are small ($+0.0027$ macro-F1, $+0.0036$ HATE-F1); since we do not run a significance test, we treat the ViHSD margin as preliminary rather than established. Accuracy *drops* on VOZ-HSD while macro-F1 and HATE-F1 rise, illustrating why accuracy is the wrong headline metric under imbalance.

6.2 Detailed per-class results

Table 5 gives the full per-class classification reports for every model on the two test sets, with macro-F1 as the headline metric. Reporting both classes separately makes the imbalance dynamics explicit: a model can post a high NON-HATE F1 (above 0.90 for every transformer) while its HATE F1—the column that actually matters—spans a much wider range, from 0.4187 to 0.7065. The proposed model's row is highlighted in bold for emphasis; it attains the best macro-F1 on both datasets (0.7819 on ViHSD, 0.8371 on VOZ-HSD).

The detailed reports localize where the macro-F1 gains come from. On VOZ-HSD the proposed model lifts HATE precision from 0.65 to 0.73 while holding HATE recall near the baseline (0.68 vs. 0.66), so the $+0.0508$ HATE-F1 gain is driven mainly by cleaner positive predictions rather than by flagging more comments. The lexical and static-embedding baselines sit at the two extremes of the precision–recall trade-off: BiLSTM + fastText reaches the highest HATE recall on both datasets (0.77 on ViHSD, 0.91 on VOZ-HSD) but at a HATE precision so low (0.38 and 0.27) that its macro-F1 is the weakest of all six models, whereas TF-IDF + SVM is conservative—high HATE precision, low HATE recall—and therefore misses most hate. The transformer models cluster much more tightly, and within that cluster the per-class view shows that ViAmpleHate's advantage is concentrated in the HATE column while its NON-HATE F1 stays essentially tied with the baseline, exactly the behaviour wanted when the minority class is the class of interest.

At the aggregate level, the confusion matrices in Figure 4 show that ViAmpleHate reduces HATE false positives relative to the baseline on ViHSD.

6.3 Effect of Vietnamese target extraction

A likely contributor to these gains is target coverage, which we define as the fraction of comments for which $T_x \neq \{0\}$ (Vietnamese NER or a target cue fires). Under English NER, only ≈0.09% of training comments contain a detected target (21/24,048), so the baseline's target-aware pathway is almost never active. With Vietnamese NER and the target-cue bank, target coverage rises to 20.0%/18.8% on the first 500 ViHSD train/validation comments and to 45.2%/43.0% on VOZ-HSD using the same 16-cue bank; attack-cue coverage is lower—5.2%/4.2% (ViHSD) and 7.6%/6.0% (VOZ). Notably, the dataset with far higher coverage (VOZ-HSD) is also where ViAmpleHate gains most (HATE-F1 $+0.0508$ vs. $+0.0036$), a suggestive cross-dataset correlation. We stress, however, that coverage is an *input* statistic: higher coverage gives relation-bank attention more to attend to but does not by itself prove that firing improves individual decisions.

6.4 Qualitative analysis

To make the error modes concrete, Table 7 shows illustrative comment types (constructed examples that mirror the observed categories; replace with real anonymized test instances for camera-ready). The pattern is that ViAmpleHate helps most when an explicit target co-occurs with an attack predicate, and least for implicit hate that names no target and uses no overt attack term.

6.5 Discussion

Three observations stand out. First, on ViHSD the change is a precision/recall reweighting rather than a uniform gain: HATE precision rises ($0.5972 \rightarrow 0.6177$) while recall *falls* ($0.6119 \rightarrow 0.5988$). Higher precision is desirable when false accusations of hate are costly, but lower recall means more hate is missed; which trade-off is appropriate depends on the deployment. We report results at a single validation-tuned threshold and leave a full precision–recall analysis to future work. On VOZ-HSD, where coverage is higher, HATE recall is also higher (0.6803; Table 6). Second, the larger VOZ-HSD gains alongside an accuracy drop indicate that the model trades majority-class accuracy for minority-class quality, which is the trade-off practitioners typically want when the minority class is the

Model	NON-HATE P	NON-HATE R	NON-HATE F1	HATE P	HATE R	HATE F1	Macro-F1	Acc.
(a) ViHSD (test)								
TF-IDF + LR	0.96	0.92	0.9382	0.48	0.62	0.5404	0.7393	0.8910
TF-IDF + SVM	0.93	0.97	0.9523	0.62	0.38	0.4739	0.7131	0.9126
BiLSTM + fastText	0.97	0.85	0.9083	0.38	0.77	0.5060	0.7072	0.8454
PhoBERT-CNN	0.96	0.92	0.9398	0.49	0.69	0.5745	0.7571	0.8945
AmpleHate (baseline)	0.96	0.95	0.9540	0.60	0.61	0.6045	0.7792	0.9175
ViAmpleHate	**0.95**	**0.96**	**0.9558**	**0.62**	**0.60**	**0.6081**	**0.7819**	**0.9205**
(b) VOZ-HSD (test)								
TF-IDF + LR	0.98	0.96	0.9708	0.48	0.73	0.5783	0.7745	0.9453
TF-IDF + SVM	0.97	0.99	0.9812	0.72	0.49	0.5850	0.7831	0.9641
BiLSTM + fastText	0.99	0.86	0.9237	0.27	0.91	0.4187	0.6712	0.8650
PhoBERT-CNN	0.98	0.98	0.9801	0.62	0.68	0.6500	0.8150	0.9623
AmpleHate (baseline)	0.98	0.98	0.9812	0.65	0.66	0.6557	0.8185	0.9643
ViAmpleHate	**0.96**	**0.97**	**0.9678**	**0.73**	**0.68**	**0.7065**	**0.8371**	**0.9420**

Table 5: Detailed per-class classification report on ViHSD (a) and VOZ-HSD (b), test sets. Macro-F1 is the primary metric and the proposed ViAmpleHate row is highlighted in bold for emphasis (not a per-column best marker). Per-class precision/recall are at the two-decimal resolution of the classifier reports; F1 and macro-F1 at four decimals.

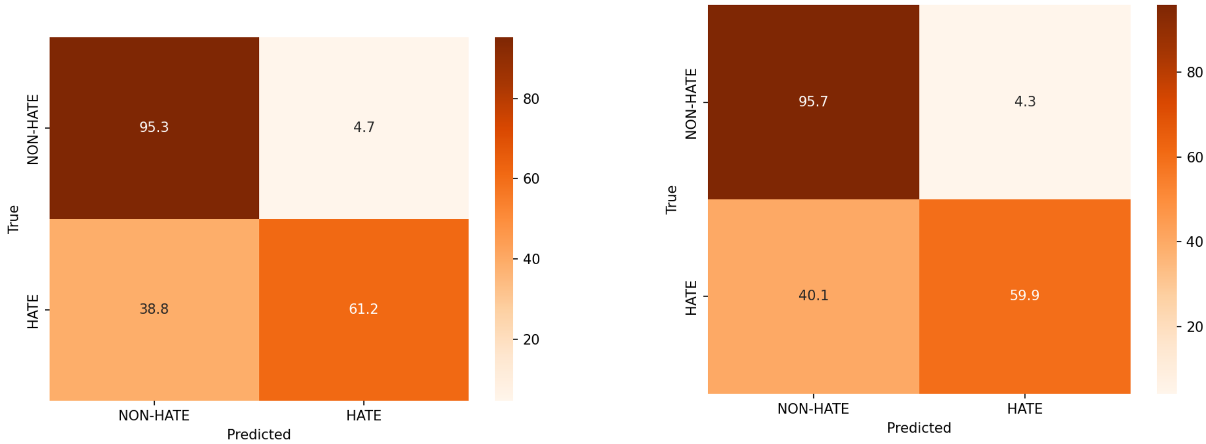

Figure 4: Confusion matrices on ViHSD: baseline AmpleHate (left) vs ViAmpleHate (right). `confusion_matrix_amplehate.png` and `confusion_matrix_viamplehate.png`; note the reduction in HATE false positives.

Data	Class	P	R	F1
ViHSD	NON-HATE	0.9541	0.9574	0.9558
ViHSD	HATE	0.6177	0.5988	0.6081
VOZ	NON-HATE	0.9638	0.9719	0.9678
VOZ	HATE	0.7347	0.6803	0.7065

Table 6: Per-class results of ViAmpleHate (test set).

target of interest. Third, the relationship between cue coverage and gains suggests a possible inexpensive path to further improvement—expanding and refining the cue banks—though this remains to be verified (Section 6.3).

From a deployment standpoint, the per-class picture in Table 5 matters more than a single head-

Comment type	Tgt	Atk	Gold
Group label + hostile predicate	yes	yes	HATE
Profanity, no group target	no	yes	NON
Sarcastic/implicit, no overt cue	no	no	HATE
Group label in neutral context	yes	no	NON

Table 7: Illustrative cases (constructed; replace with real examples). Tgt/Atk = target/attack cue present.

line number. A moderation system that wants to minimize wrongful HATE flags should weigh HATE precision heavily: there, ViAmpleHate is the strongest transformer on VOZ-HSD (0.73) and is competitive on ViHSD (0.62), while the recall-heavy BiLSTM baseline would generate far more false accusations. A system that instead wants to surface as much potential hate as possible for human review might tolerate the lower-precision, higher-recall regime of the lexical baselines, but in that case the validation-tuned threshold (Appendix A) can simply be lowered, trading precision for recall along the same model. The added components are also cheap relative to the PhoBERT backbone—three lightweight attention heads, a gate, and a one-pass cue match at preprocessing time—so the gains do not come at a meaningful inference-cost premium, and the corrected batched attention is in fact cheaper than the spurious $B \times B$ formulation it replaces.

6.6 Error analysis

We group the remaining errors into seven recurring categories. **(1) Offensive but not hate:** insults or profanity aimed at an individual rather than a protected group; over-weighting attack cues yields false positives. **(2) Implicit hate:** comments with no slur, named entity, or overt attack predicate, expressing hostility through sarcasm, stereotypes, or shared social context; cue extraction finds little to attend to. **(3) Ambiguous target reference:** pronouns and group nouns also occur in neutral or humorous comments, so a target cue without hostile context can over-predict HATE. **(4) Incomplete cue coverage:** fast-changing slang, spelling variants, abbreviations, and creative or censored profanity that a fixed cue bank cannot cover, causing false negatives. **(5) Tokenization and span-alignment errors:** NER spans, word segmentation, and PhoBERT subword tokenization may misalign—especially for multi-word expressions—so a cue can match the wrong position. **(6) Threshold sensitivity:** a validation-tuned threshold may not transfer under distribution shift. **(7) Class imbalance:** with few positives, minority-class errors persist despite weighted loss and threshold tuning. These point to expanding the cue banks via data-driven mining plus manual validation, logging per-instance predictions (with gate value, cue coverage, and confidence) for systematic false-positive/negative analysis, adding span supervision or sarcasm detection, and applying probability calibration.

7 Conclusion and Future Work

ViAmpleHate adapts AmpleHate to Vietnamese hate speech detection through Vietnamese target extraction, separate target/attack cue channels, relation-bank attention, and adaptive relation injection (plus an implementation-level batched-attention correction), trained with weighted cross-entropy plus a contrastive loss. On ViHSD and VOZ-HSD it improves macro-F1 and HATE-F1 over the AmpleHate baseline and over the TF-IDF, BiLSTM, and PhoBERT-CNN baselines, with the clearest gains on VOZ-HSD's minority class; the ViHSD improvements are small and remain to be confirmed under a controlled, multi-seed comparison. Our analysis relates these gains to target coverage, which the Vietnamese cue banks raise by roughly two orders of magnitude over English NER—a correlational link that we have not yet shown to be causal. The broader lesson is that target-awareness is useful only when adapted to the target language's surface forms, where hate targets are often informal group references rather than named entities. Future work will expand and automatically mine the cue banks with manual validation; analyse errors at the per-instance level using gate value and confidence; add span supervision, sarcasm detection, and user/discourse context; extend to multi-label and multi-target settings in the spirit of ViTHSD; and apply probability calibration for stable cross-domain deployment.

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A Hyperparameters

Parameter	Value
Encoder	`vinai/phobert-base` (768-d)
Vietnamese NER	`NlpHUST/ner-vietnamese-electra-base`
max_len	256
LR (encoder / head)	2e-5 / 5e-5
Dropout	0.1
Effective batch	32 (16 × grad-accum 2)
Epochs	up to 8 (best by val macro-F1)
α (contrastive)	0.1
Class weights w_c	inverse freq. $N/(C\,n_c)$
Contrastive margin m	0.5
Label smoothing	0.05
Seed	42
ViHSD best epoch / val F1 / thr.	4 / 0.7852 / 0.43

Table 8: Full hyperparameters.

B Cue Bank Examples

The full banks—16 target cues and 24 attack cues—are released with the code and listed in full in the accompanying report; we give ASCII-safe examples here. **Target cues** are referential (not hateful by themselves): informal group/person references, regional and demographic labels, group pronouns. **Attack cues** are hostile predicates expressing contempt, dehumanization, or parasitism (e.g., `khinh`, `an_bam`). Target and attack cues are kept in separate banks (Section 3.4).

C Reproducibility

All PhoBERT-based models share `vinai/phobert-base`; only target/relation modeling, attention computation, fusion, and loss differ between the baseline and ViAmpleHate. Best checkpoints are selected by validation macro-F1, and the HATE threshold is selected on validation and fixed for test. Cue matching is performed once during preprocessing using full token-sequence matching against the PhoBERT token stream.